Milestone 1

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Due Date: 28 August 2026

Client Requirements Analysis

1. Client background

Organisation: Thabang Building Contractors (Klerksdorp) Industry: Construction Client ID: CLI-072 Project ID: CMPG325-2026-072

Thabang Building Contractors operates from an office building in Klerksdorp, housing management, administration, design/engineering staff, sales/client liaison, and site/warehouse operations, in addition to a general contracting workforce that spends most of its time on client construction sites. The office network must support day-to-day administrative and design work, provide guest access for visiting clients, and now extend coverage to an additional floor the client has taken over.

2. Stated requirements

- Assigned addressing block: 172.30.46.0/23
- Provide appropriate connectivity and network services for the assigned scenario
- Accommodate the stated design constraint and change request (see below)
- Produce a working, testable Packet Tracer implementation
- Permit successful data exchange between the appropriate nodes
- Demonstrate the assigned networking challenge (IPv4 subnetting / VLSM)

3. Design constraint

> Management requires internet access even when the staff network is restricted.

Interpretation: the network must be able to apply internet restrictions to general staff (for example, during working hours, or as a general policy) while guaranteeing that the Management VLAN retains internet access at all times.

Design implication: this cannot be solved by VLAN separation alone,it requires an access control list (ACL) on the edge router that explicitly permits the Management subnet's traffic to the WAN interface before any deny/restriction rule that applies to other staff VLANs. See 02-network-design.md for the ACL approach.

4. Change request — CR2

> The client takes over an additional floor/area of the building and needs coverage there.

Interpretation: the existing network design must be extended to cover a new physical floor, without changing the client's assigned addressing block (172.30.46.0/23) and without disrupting the existing ground-floor VLANs.

Design implication: the VLSM plan was built with this in mind. The addressing plan allocates a dedicated subnet for the new floor from the same /23 block, and enough of the block is left unallocated to accommodate further growth if needed. See 03-ip-addressing-plan.md.

5. Identified departments / traffic groups

Based on a typical construction contractor's office structure, the following functional groups were identified for VLAN segmentation.

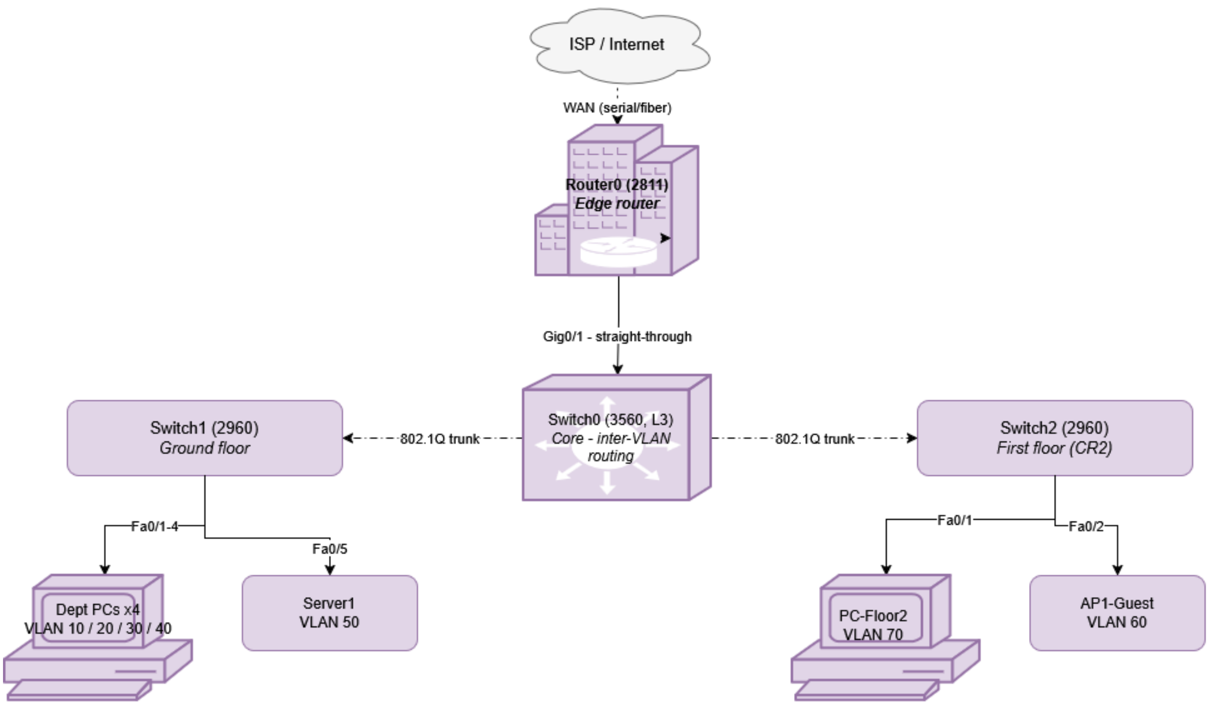
Group	Function
Management & admin	Directors, finance, HR — requires guaranteed internet access
Design & engineering	Site engineers, draughtsmen, project planning
Sales & client liaison	Client-facing sales and quoting staff
Site & warehouse operations	Equipment, materials, site coordination staff
Servers	File server / shared resources
Guest Wi-Fi	Visiting clients and contractors — restricted access
Floor 2 (CR2 expansion)	New floor taken over by the client

6. Assigned technical challenge

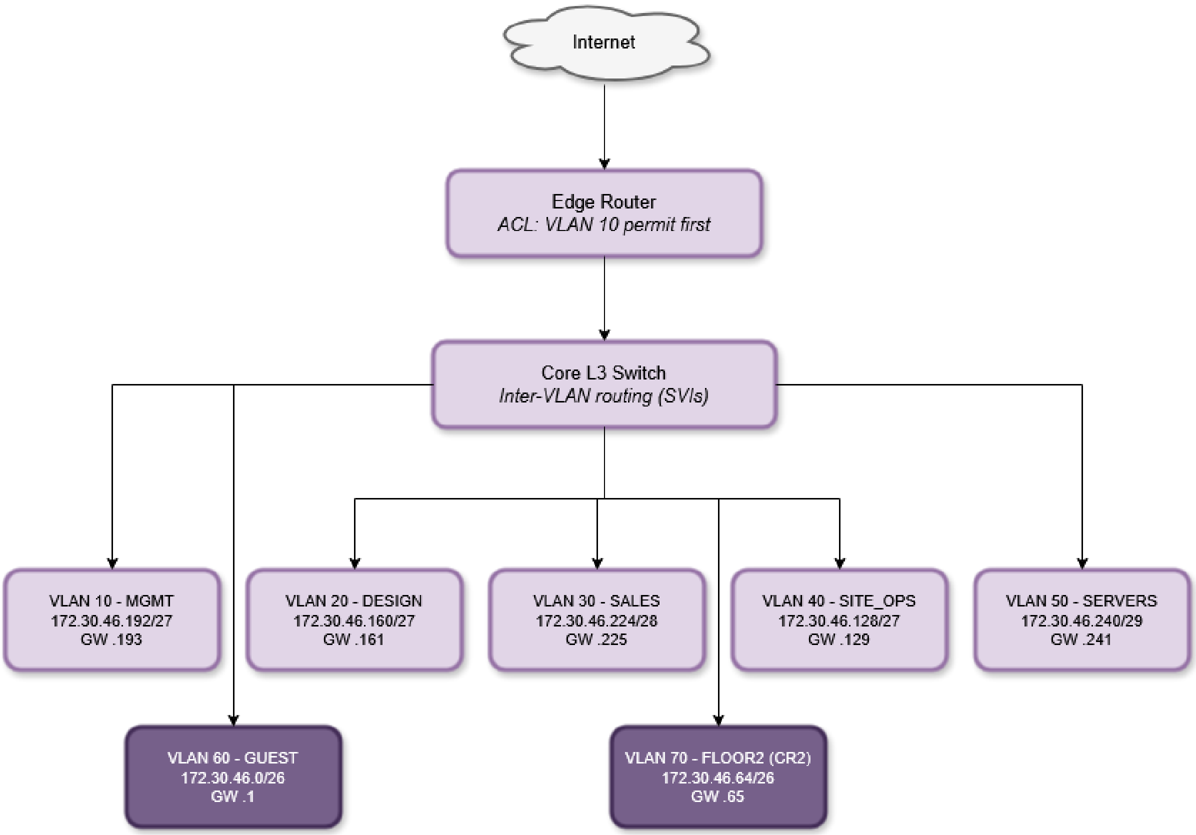
IPv4 Subnetting (VLSM addressing plan), Intermediate difficulty. The project must configure, verify, and demonstrate a VLSM addressing plan within the client network, and be able to explain what was configured, why it is appropriate, and how it was verified.

Network Design

1.Physical Topology



2.Logical Topology



1. Design goals

- Segment traffic by department/function using VLANs.
- Guarantee internet access for Management even when other staff traffic is restricted.
- Extend coverage to a new floor (CR2) without changing the assigned addressing block.
- Use VLSM to allocate address space efficiently, leaving room for future growth.
- Keep the design testable end-to-end in Cisco Packet Tracer.

2. Physical topology

Internet

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Edge Router (WAN link, extended ACL: Management always permitted)

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Core Switch (Layer 3 — inter-VLAN routing, DHCP)

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├── Ground floor switch

│ ├── Management & admin VLAN

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│     ├── Design & engineering VLAN
│     ├── Sales & client liaison VLAN
│     ├── Site & warehouse operations VLAN
│     └── Server VLAN
│
└── First floor switch  (CR2 — additional floor)
      ├── Floor 2 expansion VLAN
      └── Guest Wi-Fi (access point)
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Each floor switch trunks all its VLANs back to the core switch. The core switch (or the edge router, if a router-on-a-stick design is used instead of an L3 switch) handles inter-VLAN routing and issues DHCP leases per VLAN.

3. Logical topology / VLAN plan

VLAN numbering follows a tens-based convention grouped by function, in line with standard Cisco addressing practice: each department gets a distinct ID incrementing by 10, and VLAN 1 (the default VLAN) is left unused on all access ports as a security baseline. A dedicated native VLAN is used on all trunk links so that untagged traffic never lands on a live data VLAN.

VLAN ID	Name	Switch	Subnet	Default gateway
10	MGMT	Ground floor	172.30.46.192/27	172.30.46.193
20	DESIGN	Ground floor	172.30.46.160/27	172.30.46.161
30	SALES	Ground floor	172.30.46.224/28	172.30.46.225
40	SITE_OPS	Ground floor	172.30.46.128/27	172.30.46.129
50	SERVERS	Ground floor	172.30.46.240/29	172.30.46.241
60	GUEST	First floor	172.30.46.0/26	172.30.46.1
70	FLOOR2	First floor	172.30.46.64/26	172.30.46.65
99	NATIVE	All trunks	Not assigned to hosts	—

Notes on the convention:

- VLAN 1 is never assigned to an access port, left as the unused factory default, per standard hardening practice.
- VLAN 99 (NATIVE) carries no host traffic; it is configured as the native VLAN on every trunk port between switches and the core, so accidental untagged frames don't land on a real subnet.
- The default gateway for each VLAN is the first usable address in its subnet, and is configured as the router subinterface / switch virtual interface (SVI) address.

4. Meeting the design constraint (management always online)

Implemented as an extended ACL on the edge router's WAN-facing interface:

- permit Management subnet (172.30.46.192/27) → any destination, any time.
- deny/permit (time-based or general policy) remaining staff VLANs, per the restriction policy in effect.
- Rule order matters: the Management permit statement must be evaluated before any broader deny rule, otherwise Management traffic would be caught by the restriction too.

This should be demonstrated in Packet Tracer by testing internet reachability from a Management PC and a restricted-VLAN PC simultaneously, with the restriction active.

5. Meeting the change request (CR2 — additional floor)

Rather than requesting new address space, the first-floor expansion:

a) Uses a subnet carved from the existing 172.30.46.0/23 block (172.30.46.64/26).
b) Connects via a new access switch trunked back to the core switch, so it inherits the same inter-VLAN routing, DHCP, and ACL policy already in place.
c) Does not alter any existing ground-floor VLAN or subnet, the original design continues to function unchanged.

6. Key design decisions

- VLSM over a single flat subnet: departments have very different host counts (10–50), so VLSM avoids wasting addresses on small departments while still giving Guest Wi-Fi and the CR2 floor enough room.
- L3 core switch / router-on-a-stick: chosen to keep inter-VLAN routing centralised and to simplify ACL placement at the WAN edge.
- Guest Wi-Fi isolated to its own VLAN: keeps visiting clients off the internal departmental subnets entirely.

IP Addressing Plan (VLSM)

1. Assigned block

172.30.46.0/23 (172.30.46.0 – 172.30.47.255, 512 total addresses)

2. Method

Subnets were sized to each group's estimated host count, then allocated in descending order of size (largest first) ,the standard VLSM approach that minimises wasted address space compared with a single fixed subnet mask across all VLANs.

3. Host requirements

Group	Estimated hosts
Guest Wi-Fi	50
Floor 2 expansion (CR2)	40
Site & warehouse operations	30
Design & engineering	25
Management & admin	20
Sales & client liaison	10
Servers	6
WAN link (router–ISP)	2

4. VLSM allocation

VLAN IDs follow the tens-based convention, each subnet below is tied to a specific VLAN, not a placeholder.

VLAN ID	Group	Hosts needed	Mask	Subnet	Usable range	Broadcast
60	Guest Wi-Fi	50	/26 (62 usable)	172.30.46.0	172.30.46.1 – .62	172.30.46.63
70	Floor 2 expansion (CR2)	40	/26 (62 usable)	172.30.46.64	172.30.46.65 – .126	172.30.46.127
40	Site & warehouse ops	30	/27 (30 usable)	172.30.46.128	172.30.46.129 – .158	172.30.46.159
20	Design & engineering	25	/27 (30 usable)	172.30.46.160	172.30.46.161 – .190	172.30.46.191
10	Management & admin	20	/27 (30 usable)	172.30.46.192	172.30.46.193 – .222	172.30.46.223
30	Sales & client liaison	10	/28 (14 usable)	172.30.46.224	172.30.46.225 – .238	172.30.46.239
50	Servers	6	/29 (6 usable)	172.30.46.240	172.30.46.241 – .246	172.30.46.247
—	WAN link (router–ISP)	2	/30 (2 usable)	172.30.46.248	172.30.46.249 – .250	172.30.46.251
99	Native (trunk only, no hosts)	—	—	—	—	—

5. Unused address space

172.30.46.252/30 through the entire 172.30.47.0/24 remains unallocated, one whole spare /24 plus a /30. This is a deliberate design choice, not an oversight: it keeps the plan efficient today while leaving clear room for further growth (e.g. if the client expands beyond the CR2 floor in future), which directly supports the appropriateness argument required by the brief.

6. Why this is appropriate

- Each subnet is sized close to its actual requirement (no group is given a mask larger than it needs), which is the core justification VLSM requires over fixed-length subnetting.
- The two largest groups (Guest Wi-Fi and the CR2 floor) ,both areas expected to grow or fluctuate in headcount were given the most headroom (/26, 62 usable each).
- The WAN link uses a /30, the smallest usable subnet, since a point-to-point link only ever needs two addresses.
- Significant free space remains in the block for future subnets without needing to touch the client's assigned 172.30.46.0/23 range.

GITHUB Link

https://github.com/Koketsokgomotso/CMPG325-Project-Client-072